```
# Task1
rows = 5

for i in range(1, rows + 1):
    for j in range(i):
        print("*", end=" ")
    print()

# Task 2
rows = 4

for i in range(rows):

    for j in range(rows - i - 1):
        print(" ", end=" ")

    for k in range(2 * i + 1):
        print("*", end=" ")

    print()

# Task3
for i in range(5):
    for j in range(5):
        print("*", end="")
    print()

# Task4
n = 5

for i in range(n):
    for j in range(n):
        if i == 0 or i == n-1 or j == 0 or j == n-1:
            print("*", end="")
        else:
```

```
        print(" ", end="")
    print()

# Task5
n = 5

for i in range(n):

    for j in range(n - i - 1):
        print(" ", end="")

    for k in range(2 * i + 1):
        print("*", end="")

    print()

for i in range(n-2, -1, -1):

    for j in range(n - i - 1):
        print(" ", end="")

    for k in range(2 * i + 1):
        print("*", end="")

    print()

# Task6
n = 5
num = 1

for i in range(1, n + 1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

 # Task7

```
n = 5

for i in range(1, n + 1):

    for j in range(n - i):
        print(" ", end="")

    for j in range(1, i + 1):
        print(j, end="")

    for j in range(i - 1, 0, -1):
        print(j, end="")

    print()
```

 # Task8

```
n = 5

for i in range(n):

    print(" "*(n-i), end="")

    num = 1
    for j in range(i+1):
        print(num, end=" ")
        num = num*(i-j)//(j+1)

    print()
```

 # Task9

```
n = 5

for i in range(1, n + 1):
```

```
    for j in range(1, i + 1):

        if (i + j) % 2 == 0:
            print("1", end="")
        else:
            print("0", end="")

    print()
```

 # Task10

```
n = 5

for i in range(1, n+1):
    for j in range(1, n+1):
        print(i*j, end=" ")
    print()
```

<u>Outputs:</u>

```
(.venv) PS D:\Internship\Day28> python task1-10.py
*
* *
* * *
* * * *
* * * * *
        *
      * * *
    * * * * *
* * * * * * *
*****
*****
*****
*****
*****
*****
*   *
*   *
*   *
*****
```

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
    1
   121
  12321
 1234321
123454321
     1
    1 1
   1 2 1
  1 3 3 1
 1 4 6 4 1
1
01
101
0101
10101
1 2 3 4 5
2 4 6 8 10
3 6 9 12 15
4 8 12 16 20
5 10 15 20 25
```